



Chandigarh Engineering College Jhanjeri
Mohali-140307
Department of Artificial Intelligence (AI) and Data Sciences

IoT-Based Patient Health Monitoring System

Engineering Clinics – Multi Disciplinary Project
ECS-201

BACHELOR OF TECHNOLOGY
(Artificial Intelligence and Data Science.)



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Chapter 1

Introduction

1.1 Project Introduction

The **IoT-based Patient Health Monitoring System** is designed to continuously monitor patients' vital health parameters such as **heart rate and blood oxygen level (SpO2)**. This system is particularly useful in hospitals, clinics, and home-based healthcare to track patient health in real-time. The system uses the **MAX30100 or MAX30102 sensor** for measuring heart rate and SpO2, displays readings on an **I2C LCD**, and transmits data for further analysis using **Arduino Uno R3**.

1.2 Objectives of the Project

- To monitor **real-time heart rate and SpO2 levels** of a patient.
- To display patient health data on an **LCD** for easy visualization.
- To familiarize with **IoT concepts**, sensors, and microcontroller interfacing.
- To provide a **cost-effective, portable, and efficient** solution for health monitoring.

1.3 Tools Learned

- **Arduino Uno R3**: Programming and hardware interfacing.
- **MAX30100/MAX30102 sensor**: Pulse oximeter and heart rate sensing.
- **I2C LCD Display**: Displaying real-time health parameters.
- **Arduino IDE**: Writing and uploading code to the microcontroller.
- **Basic IoT Concepts**: Sensor data collection, real-time monitoring.

Chapter 2

System Requirements

2.1 Hardware Requirements



- Arduino Uno R3



- MAX30100 / MAX30102 Sensor



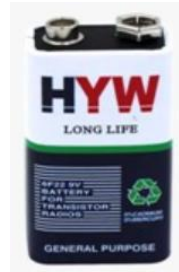
- I2C LCD Display (16x2 or 20x4)



- Breadboard and connecting wires



- USB Cable for Arduino programming



- Power supply (5V)

2.2 Software Requirements

- Arduino IDE (latest version)
- MAX30100/MAX30102 Arduino library
- I2C LCD library (LiquidCrystal_I2C)
- PC for programming and monitoring
- C++ Programming Language



Chapter 3

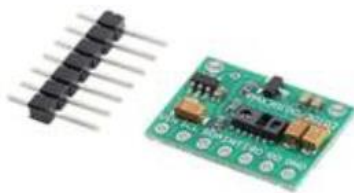
Software Requirement Analysis

3.1 Problem Definition

Traditional health monitoring requires physical presence and periodic checking. There's a growing need for real-time, continuous, and remote health monitoring, especially in critical care or elderly patients. Our system solves this by using sensors and microcontrollers to automate vital monitoring.

3.2 Module Division and Functionalities

<u>Module Name</u>	<u>Description</u>
Sensor Module	Collects heart rate and SpO2 data using MAX30100/MAX30102 sensor
Processing Unit	Arduino Uno processes sensor data and formats it
Display Module	I2C LCD module displays BPM and SpO2 values in real time
Optional Future Module	Wireless module (e.g., ESP8266) to transmit data to cloud/mobile app



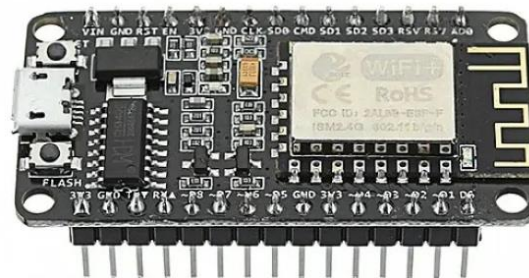
• Sensor Module



• Processing Unit



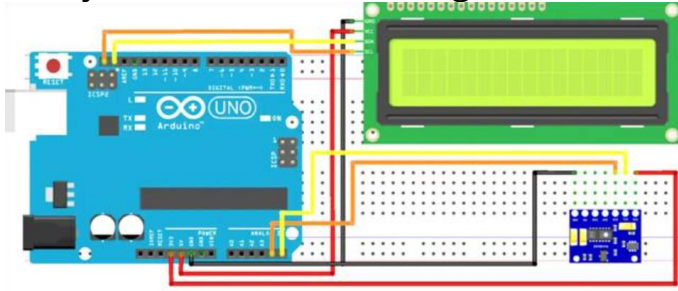
• Display Module



• Optional Future Module

Chapter 4 Software Design

4.1 System Architecture Diagram



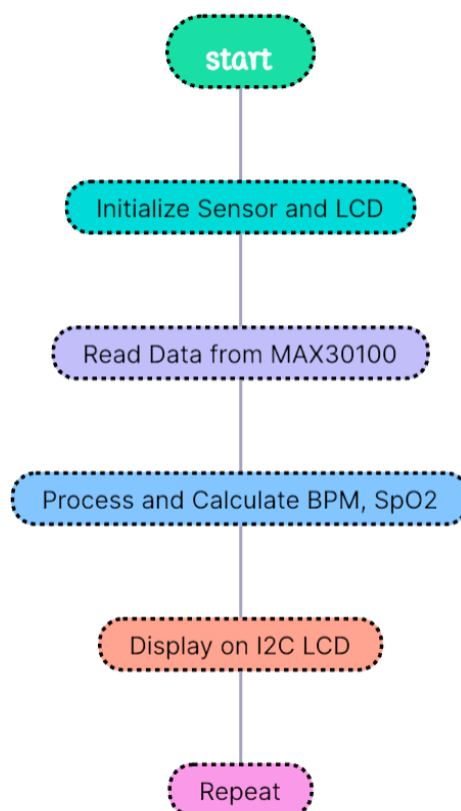
4.2 Wiring Implementation

MAX30102 Sensor to Arduino Uno:
VCC → 3.3V or 5V on Arduino
GND → GND on Arduino
SCL → A5 (or SCL pin) on Arduino
SDA → A4 (or SDA pin) on Arduino

16x2 LCD with I2C Module to Arduino Uno:

VCC → 5V on Arduino
GND → GND on Arduino
SDA → A4 (or SDA pin) on Arduino
SCL → A5 (or SCL pin) on Arduino

4.3 Flowchart (Basic Flow)





Chapter 5

Implementation

5.1 Code Outline

```
#include <Wire.h>
#include "MAX30105.h"
#include <LiquidCrystal_I2C.h>

MAX30105 sensor;
LiquidCrystal_I2C lcd(0x27,16,2);

float bpm = 0;
float spo2 = 0;

void setup() {
  lcd.init();
  lcd.backlight();
  sensor.begin(Wire, I2C_SPEED_STANDARD);
  sensor.setup(0x1F,4,2,100,411,4096);
}

void loop() {
  bpm = sensor.getHeartRate();
  spo2 = sensor.getSpO2();

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("BPM:");
  lcd.print((int)bpm);
  lcd.print("%");
  lcd.setCursor(0,1);
  lcd.print("SpO2:");
  lcd.print((int)spo2);
  lcd.print("%");
  delay(1000);
}
```

5.2 Libraries Used

- #include <Wire.h>
- #include "MAX30105.h"
- #include <LiquidCrystal_I2C.h>



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REFERENCES

- [1] **MAX30100 Pulse Oximeter Datasheet – Maxim Integrated**
- [2] Arduino Documentation – <https://www.arduino.cc>
- [3] MAX3010x Arduino Library – GitHub - https://github.com/sparkfun/SparkFun_MAX3010x_Sensor_Library
- [4] Liquid Crystal I2C Library – https://github.com/johnrickman/LiquidCrystal_I2C
- [5] YouTube Reference - <https://youtu.be/1LqBvkHTJXU?si=uALP6emBGEiecPNA>